06 — Experiments, Metrics, and Tracking

The competition is noisy. Identical files have scored 0.12–0.22 public (S34); the same byte-identical zip scored 0.10 and 0.15 (S39); a local 28/74 screen produced hidden 0.17 and 0.10 for tied configs (S35). A single score is not a measurement. This document defines the metrics, splits, and decision rules that make experiments interpretable.

The goal is not “run more experiments.” It is **make fewer, better experiments and know when a difference is real enough to spend a submission slot.**

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1. What to measure

1.1 Outcome metrics

Metric	Definition	Why
`patch_rate`	fraction of tasks with a non-empty patch	An unedited task cannot score (S5:536-537)
`resolve_rate`	fraction of tasks with `resolved=True`	The competition metric (S1)
`apply_fail_rate`	fraction of patches that fail to apply in Phase 2	Patch hygiene/format (S5:548)
`collection_fail_rate`	fraction of test runs failing at collection	Environment vs agent (S31)
`timeout_rate`	fraction of tasks hitting per-task time	Budget fit (S5:499-509)
`global_projection`	projected total time for N≈120	12h risk (S1, S29)

1.2 Process metrics

Metric	Definition	Why
`first_edit_call`	tool-call index of first source edit	Public evidence: no edit by call 12 fails 87% (S38)
`tool_calls`	calls used per task	Budget fit (S5:499-509)
`turns`	model turns per task	Loop/termination behavior
`malformed_calls`	tool calls rejected for missing/invalid args	Parser/model issue (S30, S39)
`repeat_streak`	max consecutive identical calls	Loop risk (S39)
`compaction_events`	number of compactions	Context pressure (S5:518-521)
`prompt_tokens` / `completion_tokens`	per call	Context/time cost
`adapter_effect`	output difference base vs adapter at temp 0	Adapter active (S23)
`startup_seconds`	vLLM server ready time	Startup risk (S26, S30)
`kv_cache_tokens`	`GPU KV cache size`	Long-prompt risk (S27)

1.3 Quality metrics for trajectories

Metric	Definition	Prefer
`resolved`	harness outcome	True
`efficiency`	calls/time vs budget	Lower
`style`	first edit by call 12, source-only, no test edits	Yes
`hygiene`	no scratch files, no test/config edits	Yes
`loopiness`	identical-call streak	Low
`leakage`	reads outside `/workspace`/wheels/site-packages	None

Metric	Definition	Prefer
`thinking_match`	thought tokens match inference mode	Yes
`repo_balance`	distribution across repos	Balanced

These are not all equally important. Start with outcome, patch rate, first-edit call, time, and repeat streaks.

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2. Splits

2.1 Public data is not a random sample

The 129 public tasks come from four popular repos; the hidden tasks come from private repos (S2, S31). A random public split overestimates transfer. Use **repository-grouped** holdouts where possible, and keep a set of tasks you never tune on.

2.2 Recommended splits

Split	Size	Use
Dev	40–60 tasks	Prompt/budget iteration
Holdout A	20–30 tasks	Paired comparisons; promotion decisions
Holdout B	20–30 tasks	Final confirmation; untouched until the end
Training	remaining resolved trajectories	LoRA/QLoRA
Calibration	10–15 tasks	Gold-patch/empty-patch checks

These are subsets of the 129 public training tasks, not additive. Pick disjoint task IDs for Dev/Holdout A/Holdout B; training uses trajectories from tasks outside the holdouts. If you need more tasks, reuse Dev for exploration but never for promotion.

If you have limited compute, use 20–30 holdout tasks and accept wider error bars. Do not use 2 tasks (the starter) for decisions.

2.3 Task selection

Stratify by repo and difficulty:

- Include fastapi (many tasks), rich, requests, httpx.
- Include tasks with/without hints.
- Include short and long problem statements.
- Exclude known-broken tasks from comparison, but record why.

The public 0.18 analysis found statements ≤120 chars had 7 resolves in 32 attempts vs 24 in 56 for longer statements (S39); length is a proxy, not a cause. Balance it.

2.4 Gold/empty controls

For every evaluation environment:

- Gold patch resolves on the calibration tasks.
- Empty patch fails (or the task is marked broken).
- If gold fails, fix the environment before comparing agents (S31, S32).

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3. Paired comparisons

3.1 Why paired

Run-to-run variance is large. Comparing config A on tasks 1–20 and config B on tasks 21–40 confounds config with task difficulty. Run both on the **same tasks**, same seeds, alternating order.

3.2 Design

- N = 20–40 tasks.
- Two seeds if time permits (two attempts per task).
- Alternate A/B order per task to balance drift.
- Same model revision, wheelhouse, prompt hash, eval config, GPU topology.
- Record everything in the manifest.

3.3 Analysis

For binary outcomes, use discordant pairs:

	B resolved	B failed
A resolved	both	A only
A failed	B only	neither

The sign test uses only A-only and B-only. With `d` discordant pairs and `g` wins for the candidate:

$$p = \min\left(1,\ 2\sum_{j=\max(g,d-g)}^{d}\binom{d}{j}2^{-d}\right)$$

A 6–1 win gives p≈0.125; 8–2 gives p≈0.109; 9–1 gives p≈0.021. Small samples cannot establish small effects. Report the raw counts, not just a p-value.

For continuous metrics (time, calls), report median and p95, and the paired difference per task. A 10% mean improvement driven by one task is not a win.

3.4 Practical decision rule

Promote a candidate only if:

1. It has **≥8 discordant pairs and ≥75% wins** (or two seeds) on the paired holdout, **and**
2. It does not increase p95 time, setup time, or timeout rate materially, **and**
3. It does not reduce patch rate, **and**
4. The change is mechanistically plausible, **and**
5. It has a hidden score before it is eligible for a final slot.

At 6 discordant pairs, 2 wins gives p≈0.69; at 8, 2 wins gives p≈0.29; at 10, 2 wins gives p≈0.11. Report the raw discordant counts, not just the net. A 3+ task win with better p95/patch rate and a hidden score is a real candidate.

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4. Run manifest

Every run must be reproducible. Save a JSON file next to the results:

```
{
  "run_id": "2026-10-24-baseline-01",
  "date_utc": "2026-10-24T12:00:00Z",
  "competition": "gemma-4-developer-agent",
  "purpose": "baseline prompt-only on holdout A",
  "agent_dir": "submission/",
  "agent_zip_sha256": "...",
  "model": "gemma-4-31b-it-qat-w4a16-ct",
  "model_revision": "...",
  "vllm_version": "0.19.1",
  "vllm_wheel_sha256": "...",
  "wheelhouse_hash": "...",
  "python_version": "3.12.x",
  "cuda_version": "...",
  "gpu": "4x A100 40GB",
  "sandbox": "docker",
  "eval_config": {
    "timeout_seconds": 240,
    "max_tool_calls": 28,
    "max_time_minutes": 8,
    "max_turns": 80
  },
  "sampling": {
    "temperature": 0.2,
    "top_p": 0.95,
    "max_output_tokens": 8192,
    "thinking_budget": 4096,
    "include_thoughts": true
  },
  "prompt_sha256": "...",
  "tool_schema_sha256": "...",
  "adapter": null,
  "adapter_sha256": null,
  "adapter_rank": null,
  "adapter_alpha": null,
  "task_ids": ["..."],
  "seeds": [42],
  "git_commit": "...",
  "notes": "first baseline; no adapter"
}
```

For training runs, add:

```
{
  "training_base": "google/gemma-4-31B",
  "training_base_revision": "...",
  "training_base_note": "use the verified BF16 ID or a verified QAT-
unquantized ID/revision",
  "dataset_sha256": "...",
  "num_trajectories": 0,
  "num_samples": 0,
  "sequence_len": 8192,
```

```
  "rank": 16,
  "alpha": 16,
  "target_modules": "...",
  "learning_rate": 5e-5,
  "epochs": 1,
  "batch_size": 1,
  "grad_accum": 8,
  "optimizer": "adamw_8bit",
  "train_loss": 0.0,
  "val_loss": 0.0,
  "tokens_per_second": 0.0,
  "peak_vram_gb": 0.0,
  "gpu_hours": 0.0
}
```

Hash the prompt/config/zip. If you can't reproduce the hash, you can't reproduce the run.

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5. Results schema

Use JSONL for per-task results:

```
{"run_id":"...","task_id":"fastapi_15661","repo":"fastapi/fastapi","resolved
":false,"patch_size":0,"files_changed":0,"tool_calls":10,"turns":8,"duration
_s":70.2,"setup_s":55.0,"first_edit_call":null,"malformed_calls":0,"max_repe
at_streak":3,"compaction_events":0,"error":"BudgetExceeded"}
```

Aggregate:

```
{"run_id":"...","n":33,"resolved":9,"patch_rate":0.82,"apply_fail":0,"collec
tion_fail":1,"timeout":2,"p50_s":210,"p95_s":480,"mean_calls":22.1,"mean_tur
ns":18.4,"malformed_rate":0.04,"global_projection_h":9.8}
```

Keep the raw traces and patches; the aggregate is not enough to debug.

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6. Experiment backlog

Run these in order. Do not skip the baseline.

6.1 Scaffold experiments

ID	Change	Hypothesis	Metric
E1	Reproduce public prompt-only config (author-reported 0.24)	Baseline works	resolve, patch rate, time
E2	5 vs 6 vs 8 min cap	Lower cap improves global completion without losing solves	resolve, timeout, projection
E3	24 vs 28 vs 32 calls	Fewer calls reduces loops; more calls helps hard tasks	resolve, patch rate, calls
E4	thinking 0/512/2048/4096	Bounded thinking reduces loops; too much costs time	resolve, malformed, time
E5	output 8192 vs 12288/16384	Higher cap reduces truncation but hurts prompt room	truncation, resolve
E6	graph tools on/off	Graph tools help localization or waste calls	first edit, resolve
E7	edit-by-call 10/12/14	Earlier edits increase patches but may increase wrong edits	patch rate, resolve
E8	read-only explorer sub-agent	Context isolation improves hard tasks	resolve, time, calls
E9	prompt length 2k/4k/6k	Shorter prompt survives compaction better	resolve, compaction
E10	temp 0.2 vs 0.7	Lower temp reduces loops; higher explores	repeat streak, resolve

6.2 Data/training experiments

ID	Change	Hypothesis	Metric
T1	10-step smoke test	Fits GPU, loss finite	VRAM, tokens/s

ID	Change	Hypothesis	Metric
T2	31B QLoRA rank 8 vs 16	Lower rank more stable	loss, adapter effect, resolve
T3	alpha 8/16 at inference (rank 16); 32 diagnostic	Scale 1.0 best; 2.0 degenerate	loops, time, resolve
T4	all-layer vs last-N coverage	Partial coverage more stable on W4A16	reproducibility, resolve
T5	thinking vs no-thinking data	Match inference	thought length, time, resolve
T6	300 vs 1,000 trajectories	More data helps until quality drops	val loss, resolve
T7	quality-filtered vs all resolved	Quality beats quantity	resolve
T8	SRFT step masking	Unresolved data helps	resolve
T9	31B vs 12B adapter	31B better despite speed	resolve, time
T10	E4B LoRA	Small-model bonus	small-model score

Do not run T1–T10 in parallel with E1–E10 on the same GPU without a queue; they compete for memory. Use separate GPUs or sequential runs.

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7. Local-to-hidden calibration

You cannot fully calibrate local to hidden, but you can track the relationship:

- For each hidden submission, record the local holdout resolve rate and the hidden score.
- Plot local vs hidden; compute the ratio and residual.
- Do not fit a conversion factor on 2–3 points. Use it as a warning, not a prediction.
- If local improves but hidden does not across 2–3 submissions, your local set is not representative; change the holdout/repos.

Public evidence: local 28/74 tied configs scored 0.17 and 0.10 hidden (S35); local wins did not always transfer (S39). Treat local as a screening tool, not an oracle.

8. Decision log

Keep a short decision log:

```
2026-10-24 — E1 baseline: 9/33, patch 82%, p95 480s. Projected 9.8h. Submit.
2026-10-26 — E2 5min: 8/33, patch 80%, p95 360s. Projected 8.2h. Not
promoted (1-task loss, but time better; keep as fallback).
2026-10-28 — E4 thinking512: 11/33, patch 85%, p95 420s. Promoted for next
submission.
2026-10-30 — T2 rank16 adapter: 12/33 vs base 9/33, but p95 560s. Alpha
sweep needed.
```

The log prevents re-litigating the same experiment and preserves the reasoning when context is lost.

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9. Statistical cautions

- **Multiple comparisons:** if you test 10 prompt variants on 20 tasks, one will look good by chance. Pre-register the main metric and limit variants.
- **Survivorship:** tasks that fail early may be dropped by the harness; don't compare only the tasks that produced patches.
- **Correlated tasks:** tasks from the same repo share code/conventions; treat repo as a cluster, not independent samples.
- **Seed variance:** one seed is not a distribution. Two seeds on 20 tasks is better than one seed on 40; alternate order.
- **Display truncation:** public scores are two decimals; one public-LB task is ~1/58 ≈ 0.017, one final-split task is ~1/60 ≈ 0.017, and the 0.01 display step is smaller than one task (S4, S39). Do not overinterpret a 0.01 difference.
- **Global time coupling:** a config that solves more tasks but times out globally has zero utility. Always report completion risk with quality (S29, S39).

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10. Experiment checklist

- ☐ Run ID and manifest saved.

- ☐ Prompt/config/zip hashes recorded.
- ☐ Model/adapter revisions recorded.
- ☐ Holdout task IDs fixed; no training overlap.
- ☐ Gold/empty controls pass.
- ☐ Same tasks/seeds/order for A/B.
- ☐ Raw traces/patches preserved.
- ☐ Per-task JSONL + aggregate.
- ☐ Decision and rationale logged.
- ☐ Known-good bundle updated only after a win.
- ☐ Submission slot not wasted on a 1-task local difference.